

NAVLE Review

Equine Study Guide

A complete review of equine medicine,
organized by body system.

Cardiology · Respiratory & Upper Airway · Colic & GI
Liver, Muscle & Endocrine · Neurology · Lameness & the Foot
Ophthalmology · Dermatology · Infectious Disease
Parasitology · Reproduction & Neonatology · Toxicology
Pharmacology & Anesthesia

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HOW TO USE THIS GUIDE

Yellow boxes are board pearls and buzzword-to-diagnosis shortcuts. **Blue boxes** are look-alike discriminators — the "how do I tell these two apart" questions the NAVLE builds whole items around. **Red boxes** flag genuine clinical dangers.

R = reportable **FAD** = foreign animal disease **Z** = zoonotic

1 | Cardiology

Topic	Key Facts
Most common arrhythmia overall	Atrial fibrillation is the most common pathologic arrhythmia of horses; 2nd degree AV block is the most common arrhythmia at rest and is physiologic in the resting horse with a low heart rate — it disappears with exercise or excitement. Mobitz Type I (Wenckebach) — progressive PR/PQ prolongation before a dropped beat; the normal equine finding. Mobitz Type II — AV nodal, no progressive prolongation; pathologic.
Atrial fibrillation — recognition	Irregularly irregular rhythm that does NOT disappear with exercise — that is the key discriminator from physiologic 2° AV block. No P waves; irregular R-R intervals. Confirm with an ECG.
Atrial fibrillation — treatment & prognosis	Quinidine (a class IA sodium channel blocker) for chemical conversion; add digoxin for rate control if quinidine produces an accelerated ventricular response. Transvenous electrical cardioversion is the modern alternative. Prognosis is GOOD with cardioversion, provided there is no underlying structural heart disease.
Aortic rupture	The FRIESIAN — older male. A breed-specific predisposition worth memorizing on its own.
Holodiastolic musical decrescendo murmur	Aortic insufficiency/regurgitation — the classic "dove coo" murmur of the older horse. Usually well tolerated.
Thumps (synchronous diaphragmatic flutter)	Associated with HYPOCALCEMIA (and other electrolyte derangements) — the phrenic nerve depolarizes alongside the atria, so the flank twitches in time with the heartbeat. Endurance horses after prolonged exertion and sweating.
Epinephrine release	Increased PCV/HCT from splenic contraction — the horse stores a huge red cell reserve in the spleen. An excited horse has an artifactually high PCV.
DIC treatment	Heparin. DIC is a state of excessive clotting; aminocaproic acid does the opposite — it prevents breakdown of a formed clot and is used to control excessive bleeding.

2 | Respiratory & Upper Airway

2.1 Equine Asthma (RAO / Heaves)

Aspect	Detail
Mild-to-moderate signs	Exercise intolerance / poor performance and chronic cough. No signs at rest.
Severe (heaves) — classic findings	Mild neutrophilia · hyperfibrinogenemia · increased SAA · BAL containing CURSCHMANN SPIRALS (inspissated mucus plugs). Also a "heave line" from abdominal effort, nostril flare, and increased expiratory effort.
Diagnosis	BAL — Curschmann spirals and neutrophilia. Environmental history (hay, indoor stabling, or summer pasture for the pasture-associated form).
The confirmatory drug test	Buscopan (N-butylscopolammonium) — a bronchodilator; improvement after administration confirms reversible bronchoconstriction, i.e. asthma.
Treatment	Clenbuterol (beta-2 agonist bronchodilator) and dexamethasone (corticosteroid). Environmental management is the real treatment — soaked hay or complete feed, turnout, dust reduction.

2.2 Upper Airway Surgical Disease

Condition	Presentation	Treatment
Laryngeal hemiplegia ("roaring")	Left recurrent laryngeal neuropathy → failure of arytenoid abduction → inspiratory noise and exercise intolerance. Large breeds.	Prosthetic laryngoplasty ("tie-back") of the affected side.
Dorsal displacement of the soft palate (DDSP)	Expiratory "gurgling" noise, choking-down during exercise.	Tie-forward procedure, or staphylectomy (resection of the caudal free margin of the soft palate).
Epiglottic entrapment	Abnormal airway noise; the aryepiglottic fold envelops the epiglottis.	Surgical transection of the entrapping fold.
Ethmoid hematoma	UNILATERAL EPISTAXIS — the buzzword — that is intermittent and not exercise-associated.	Intralesional formaldehyde via endoscope. Perform CT or MRI FIRST to assess the cribriform plate — formaldehyde reaching the cribriform plate causes fatal neurologic injury.
Guttural pouch tympany	Young horse with a non-painful, air-filled swelling in the parotid region.	Surgical fenestration of the median septum.
Temporohyoid osteoarthropathy (THO)	Arthritis at the junction of the stylohyoid bone and the temporal bone. Causes CN VII and VIII deficits — head shaking, facial paralysis, vestibular signs.	Unilateral ceratohyoidectomy to reduce leverage on the fused joint.
Viborg's triangle	The surgical landmark for guttural pouch access, bounded by three structures: 1. Ramus (bend) of the mandible 2. Linguofacial vein 3. Tendon of the sternocephalicus muscle	Know the three borders — a classic anatomy question.

2.3 Lower Airway & Pleural Disease

Condition	Key Facts
Pleuropneumonia	Main organism: <i>Streptococcus equi</i> subsp. <i>zooepidemicus</i> . Classic history is long-distance transport with the head tied up ("shipping fever").

Condition	Key Facts
Acute bronchointerstitial pneumonia	Two risk factors: heat stress and recent antimicrobial use . Foals; high mortality.
EIPH (exercise-induced pulmonary hemorrhage)	Diagnosed by hemosiderin within alveolar macrophages ("siderophages") on BAL.
EHV-5	Equine multinodular pulmonary fibrosis (EMPF) — progressive nodular interstitial fibrosis in adult horses.
Dysphagia	Most common complication: aspiration pneumonia .
Anhidrosis	Diagnose with the terbutaline sweat test — intradermal beta-2 agonist fails to provoke sweating.
Why horses cannot vomit	A strong lower esophageal sphincter with an oblique, low angle of entry into the stomach — functionally a one-way valve. This is why gastric reflux must be relieved by nasogastric tube and why gastric rupture is a real risk in obstructive colic.

3 | Colic & Gastrointestinal Disease

3.1 Functional Anatomy You Will Be Asked About

Structure	Detail
Cecum	Sits on the RIGHT side.
Small colon	The segment with ONE broad antimesenteric band — and the site where fecaliths lodge. (The right ventral colon has 4 bands .)
Most common impaction site	PELVIC FLEXURE of the large colon — where the diameter narrows sharply. Second most common: the junction of the right dorsal colon with the transverse colon .
Epiploic foramen	The opening into the omental bursa, bounded by three structures: 1. Caudate lobe of the liver 2. Portal vein 3. Caudal vena cava Entrapment: the distal jejunum and ileum become incarcerated through the foramen — a strangulating lesion requiring emergency surgery. The behavioral association: CRIBBING.
Normal gastric capacity	8–10 L — small relative to body size, which is why gastric rupture occurs when reflux is not relieved.

THE CRIBBING QUESTION

Cribber + acute severe colic + small intestinal distension → EPIPLOIC FORAMEN ENTRAPMENT. This is one of the tightest signalment-to-lesion links in equine medicine.

3.2 Gastric Ulcer Syndrome (EGUS)

Type	Features	Treatment
Equine SQUAMOUS gastric disease	The upper , non-glandular mucosa. Graded 0–4 . Caused by acid splashing during exercise and intermittent feeding.	Responds well to a 28-day course of omeprazole. Feeding alfalfa is useful — it buffers and raises stomach pH.
Equine GLANDULAR gastric disease	The lower , glandular mucosa. Described morphologically as hemorrhagic, fibrinosuppurative, raised, or flat . Reflects a breakdown of mucosal defense rather than acid excess — harder to treat.	Omeprazole PLUS sucralfate PLUS misoprostol. Longer courses; often needs management change.
What divides them	The MARGO PLICATUS — the abrupt line separating squamous from glandular mucosa. Naming it is the exam question.	

3.3 Colic Lesions & Enteropathies

Condition	Features	Management
Large colon impaction	Pelvic flexure; slow-onset, mild-to-moderate pain, decreased manure.	Initial treatment: 6–8 L of water by nasogastric tube every 4–6 hours , plus analgesia and IV fluids. Withhold feed.
Enterolith	Composed of MAGNESIUM AMMONIUM PHOSPHATE (struvite) , layered around a nidus. Alfalfa-heavy diets and California are classic associations.	Diagnosis: abdominal radiographs — about 80% sensitive. Treatment: midline celiotomy to remove them.

Condition	Features	Management
Duodenitis–proximal jejunitis (DPJ)	Large-volume gastric reflux, fever, depression. Associated with laminitis and is itself a proximal enteritis .	Medical: repeated nasogastric decompression, IV fluids, anti-endotoxic therapy. Distinguish from strangulating obstruction, which needs surgery.
Colic with large volumes of gastric reflux	Points to a proximal (small intestinal) lesion — DPJ, ileal impaction, or a strangulating small intestinal obstruction.	Decompress and determine surgical vs. medical.
Equine proliferative enteropathy	<i>Lawsonia intracellularis</i> — weanling and yearling horses. Causes diarrhea and profound HYPOPROTEINEMIA with ventral edema.	Tetracyclines (oxytetracycline or doxycycline); also erythromycin/rifampin. Plasma for severe hypoproteinemia.
Most common IBD in horses	Lymphocytic-plasmacytic enterocolitis.	Corticosteroids.
Most common cause of peritonitis in an adult horse	Large or small strongyle migration (cyathostomiasis). Think <i>Actinobacillus equuli</i> as the organism.	Antimicrobials, peritoneal lavage, anthelmintic control.
Rectal prolapse	Common cause: dystocia .	Reduce, purse-string, treat the cause.
Rectal tears — the four grades	Grade 1 — mucosa and submucosa only Grade 2 — muscularis alone is ruptured (mucosa intact) Grade 3 — mucosa, submucosa, and muscularis (serosa intact) Grade 4 — full-thickness tear into the abdomen	Grade 4 is a surgical emergency and a malpractice-exposure event. Stop the exam, sedate, give antibiotics and anti-inflammatories, and refer immediately.
Prerenal azotemia	BUN:creatinine ratio over 20:1.	Fluid therapy.

3.4 Infectious Diarrhea

Agent	Presentation	Diagnosis & Treatment
Potomac Horse Fever (equine monocytic ehrlichiosis)	<i>Neorickettsia risticii</i> — an intracellular bacterium infecting MONOCYTES . Fever, diarrhea, laminitis , abortion. Seen in the FALL on the EAST COAST , and generally near water — the organism lives in trematodes in aquatic insects (caddisflies/mayflies), which horses ingest accidentally.	Blood or fecal PCR. Treat with OXYTETRACYCLINE.
Rotavirus	The classic stem: a 2–5 month old foal with profuse "cow pie" diarrhea and dehydration. Destroys enterocytes → lactase deficiency → maldigestion and malabsorption.	Resistant to PHENOLIC disinfectants — use bleach or accelerated hydrogen peroxide. Vaccinate mares in late gestation. Supportive care.
<i>Clostridium perfringens</i> type C	Fulminant HEMORRHAGIC diarrhea in foals under 10 days of age. Peracute and often fatal.	Metronidazole for clostridial enterocolitis in foals; antitoxin, plasma, and aggressive supportive care.
<i>Clostridioides difficile</i>	Antimicrobial-associated colitis. The classic setup: a foal is being treated with erythromycin for <i>Rhodococcus</i>, and the MARE develops enterocolitis from grooming and ingesting the drug.	Metronidazole.

ERYTHROMYCIN, THE FOAL, AND THE MARE

This is a favorite exam item because it tests two facts at once: **macrolides are the treatment for *Rhodococcus equi* in the foal**, and **macrolides cause fatal colitis in the adult horse**. The mare is poisoned by her own foal's medication. Warn owners to wear gloves and prevent the mare from licking the foal's mouth after dosing.

4 | Liver, Muscle & Endocrine

4.1 Hepatic Disease

Condition	Key Facts
Liver-specific enzyme	SDH (sorbitol dehydrogenase) is the liver-specific enzyme in the horse (and cow, sheep, goat). GGT is the useful cholestatic marker. ALT is not liver-specific in large animals.
Pyrrolizidine alkaloid (PA) toxicosis	Hepatocytes cannot divide → progressive hepatic failure and hepatic encephalopathy. Three diagnostic findings: MEGALOCYTOSIS · PERIPORTAL FIBROSIS · BILE DUCT HYPERPLASIA. Plants: ragwort (<i>Senecio</i>), fiddleneck (<i>Amsinckia</i>), groundsel, tansy.
Theiler's disease (serum hepatitis)	The look-alike for PA toxicosis in horses. Acute hepatic necrosis weeks after administration of an equine biologic (tetanus antitoxin, plasma). Now attributed to equine parvovirus-hepatitis. Differentiate from PA toxicity by histopathology — PA gives megalocytosis and fibrosis; Theiler's gives acute centrilobular necrosis without megalocytes.
Secondary photosensitization	Phylloerythrin — the normal chlorophyll breakdown product — is normally cleared by the liver . When the liver is damaged it accumulates in the skin and reacts with UV light, causing burns on white/unpigmented areas . Primary photosensitization (no liver disease) comes from St. John's wort, <i>Erodium</i>, brassicas, and <i>Hypericum</i> — and from tetracyclines .
Portosystemic shunt in FOALS	Two features: neurologic signs , and — the trap — NORMAL liver enzymes . A shunt bypasses the liver rather than damaging it.
Hemochromatosis	An iron storage disease with hemosiderin deposited in hepatocytes . Related: oral IRON supplementation causes liver injury and hepatic encephalopathy in FOALS — never give oral iron to a neonatal foal.
SAA vs. fibrinogen	Both are made by the LIVER . SAA rises and falls faster than fibrinogen , making it the better marker of acute and resolving inflammation.

4.2 Muscle Disease

Condition	Pathophysiology & Signs	Management
HYPP (hyperkalemic periodic paralysis)	Autosomal DOMINANT mutation of the voltage-gated sodium channel of skeletal muscle. Channels fail to close , sodium leaks in, the muscle stays depolarized and cannot repolarize → weakness and paralysis , with potassium leaving the cell → HYPERKALEMIA . Quarter Horses tracing to Impressive.	Three-part emergency treatment: 1. 5% sodium bicarbonate IV — drives K^+ into the cell by exchanging for H^+ 2. 5% dextrose IV ($\pm$ insulin) — drives K^+ intracellularly 3. Calcium borogluconate IV — stabilizes the myocardium against hyperkalemia Long term: low-potassium diet, acetazolamide.
Exertional rhabdomyolysis ("tying up")	Two lab indicators: elevated CK and AST . CK peaks early; AST stays up longer.	Rest, fluids (myoglobin is nephrotoxic), NSAIDs.
PSSM type 1	Polysaccharide storage myopathy — abnormal glycogen accumulation.	Diet HIGH IN FAT and LOW IN STARCH , plus consistent daily exercise.
Post-anesthetic myositis	Compartment-syndrome-like muscle injury after recumbency.	Dantrolene — a ryanodine receptor antagonist and muscle relaxant.

4.3 Endocrine

Condition	Pathophysiology	Diagnosis & Treatment
PPID (pituitary pars intermedia dysfunction, "equine Cushing's")	LOSS OF DOPAMINE production by hypothalamic neurons that normally inhibit the pars intermedia → unrestrained POMC-derived hormone output. Hirsutism/hypertrichosis, laminitis, muscle wasting, PU/PD.	Diagnose with a resting ACTH (accounting for the seasonal autumn rise); TRH stimulation improves sensitivity. Treat with PERGOLIDE — a DOPAMINE AGONIST that replaces the missing inhibition.
EMS (equine metabolic syndrome)	Insulin dysregulation with regional adiposity and laminitis risk.	Test a resting insulin first, then an oral sugar test (OST) if resting insulin is equivocal. Diet and exercise; levothyroxine or metformin as adjuncts.
Nutritional secondary hyperparathyroidism ("big head" / bran disease)	Excess dietary phosphorus (or oxalate-rich grasses) blocks calcium absorption → PTH rises → fibrous osteodystrophy of the facial bones.	Bloodwork: increased PTH · HYPERPHOSPHATEMIA · normal-to-low calcium. Correct the calcium:phosphorus ratio.

RAPID-FIRE GI, LIVER & METABOLIC RECALL

Cribber with acute colic → **epiploic foramen entrapment**.

Enterolith → **magnesium ammonium phosphate** → radiographs 80% sensitive → surgery.

Squamous vs. glandular ulcers divided by the **margo plicatus**.

Foal with cow-pie diarrhea → **rotavirus**, resistant to **phenolics**.

Fall, east coast, near water, fever + diarrhea + laminitis → **Potomac horse fever** → **oxytetracycline**.

Weanling with diarrhea and severe hypoproteinemia → **Lawsonia** → tetracyclines.

Megalocytosis + bile duct hyperplasia → **pyrrolizidine alkaloids**.

Foal with neuro signs and **normal** liver enzymes → **portosystemic shunt**.

QH that goes down with hyperkalemia → **HYPP** → bicarb, dextrose, calcium.

Hairy coat that won't shed + laminitis → **PPID** → **pergolide**.

5 | Neurology

THE ATAXIC HORSE — THE SINGLE HIGHEST-YIELD EQUINE NEUROLOGY ITEM

Five conditions produce an ataxic horse and the exam separates them by **age, symmetry, and one diagnostic test**. Work the table in §5.1 before anything else.

5.1 The Ataxic Horse Algorithm

Disease	Age	Signs	Diagnosis & Treatment
Cervical vertebral malformation (Wobblers / CVM)	Type 1: young Type 2: older	SYMMETRIC ataxia, WORSE IN THE PELVIC LIMBS, UMN signs, and circumduction of the hindlimbs.	Myelography with contrast. See the two types below.
EPM	Any	ASYMMETRIC ataxia with focal muscle atrophy — the discriminator from wobblers. Cranial nerve deficits possible. <i>Sarcocystis neurona</i> ; opossum definitive host.	Check a CBC for anemia before and during treatment. Three treatment options: antifolate antibiotics (sulfadiazine/sulfamethoxazole + pyrimethamine, 6 months); oral ponazuril, 28 days ; diclazuril pellets, 28 days .
EDM (equine degenerative myeloencephalopathy)	Under 1 year	SYMMETRIC ataxia in a young horse.	Caused by LOW VITAMIN E — foals lacking green feed or vitamin E supplementation. Supplement vitamin E.
EHV-1 myeloencephalopathy	Adult	ASCENDING PARALYSIS and POSTERIOR PARESIS ; urinary incontinence and a flaccid tail. Often a traveling horse with a fever.	CSF: XANTHOCHROMIC (yellow, from prior hemorrhage) with elevated protein. Vasculitis and thrombosis of spinal vessels. Supportive care; strict biosecurity.
Cauda equina syndrome (polyneuritis equi)	Adult	Poor tail tone and fecal incontinence ; perineal analgesia; urinary retention.	An autoimmune reaction against myelin P2 protein. Prognosis: POOR.
Equine motor neuron disease (EMND)	Adult	Three signs: weight loss with a GOOD appetite, inability to stand / constant weight shifting, and generalized muscle atrophy. Trembling, low head carriage. Not truly ataxic — weak.	Caused by LOW VITAMIN E. Diagnose by biopsy of the sacrocaudalis dorsalis (tail head) muscle and the spinal accessory nerve.

The two types of Wobblers

Type	Signalment	Mechanism & Site	Treatment
Type 1 Dynamic compression	Young horse — Thoroughbred	Compression occurs when the neck is FLEXED . Most common at C3–C4 and C4–C5. Associated with OCD and rapid growth.	Dietary restriction and limited exercise — slow the growth rate.
Type 2 Static compression	Older horse — Warmblood	Osteophytes (articular process osteoarthritis) compress the cord regardless of position. Most common at C5–C6 and C6–C7.	NSAIDs and stall rest ; surgical stabilization in selected cases.

WOBLERS VS. EPM — THE DISCRIMINATOR THAT ANSWERS MOST QUESTIONS

WOBLERS = SYMMETRIC ataxia, worse behind, no muscle atrophy, young growing horse → **myelogram**.

EPM = ASYMMETRIC ataxia with **focal muscle atrophy** and possible cranial nerve signs → antiprotozoal therapy.

Symmetry is the fastest discriminator on the exam floor and on paper.

5.2 Encephalitides

Disease	Virus	Features	Diagnosis
EEE and WEE	Togavirus (alphavirus)	Mosquito-borne; birds are the reservoir. EEE has very high mortality.	IgM capture ELISA — the test of choice for all equine arboviral encephalitides. CSF in EEE: NEUTROPHILIC pleocytosis with increased protein — unusual for a viral encephalitis and therefore a testable detail.
VEE R	Alphavirus	High viremia — which is why the horse is an amplifying host for VEE (unlike EEE/WEE, where horses are dead-end hosts). Reportable.	IgM ELISA; report.
West Nile virus	Flavivirus	Two signs to name: FINE MUSCLE TREMORS and HYPERESTHESIA . Also ataxia and cranial nerve deficits.	IgM ELISA . Vaccinate — highly effective.
Borna disease	Bornavirus	Associated with Joest-Degen intranuclear inclusion bodies .	Histopathology.
Hendra virus FAD Z	Henipavirus (paramyxovirus)	Fatal respiratory and neurologic disease in horses in AUSTRALIA; endemic in FRUIT BATS . Highly fatal zoonosis — has killed veterinarians.	Exclusion diagnosis in any Australian horse with fever plus respiratory or neurologic signs. Extreme PPE.
Rabies R Z	Lyssavirus	Protean — can mimic colic, lameness, or any neurologic presentation.	Always on the differential for an unexplained equine neurologic case. Vaccinate.

5.3 Neuromuscular & Other

Condition	Features	Treatment
Botulism — adult	Flaccid paralysis from blockade of acetylcholine release at the neuromuscular junction. Dysphagia, weak tongue tone, trembling, recumbency.	Antitoxin; supportive care and ventilation. Vaccinate in endemic (Kentucky, mid-Atlantic) regions.
Shaker foal syndrome	Botulism in foals 2 weeks to 8 months. Different from adult botulism — foals ingest the SPORE , which then germinates and produces toxin IN the gut (a toxico-infection), rather than ingesting preformed toxin. <i>C. botulinum</i> type B.	Antitoxin and potassium penicillin (to kill the vegetative organisms in the gut). Nursing care.
Tetanus	<i>Clostridium tetani</i> ; blocks inhibitory neurotransmitter release → spastic paralysis, third eyelid prolapse, "sawhorse" stance, lockjaw.	Three-part treatment: 1. Antitoxin IV 2. Metronidazole PO or per rectum 3. Penicillin Plus a dark quiet stall, sedation, and wound debridement.
Peripheral vestibular disease — four signs	1. Head tilt 2. Circling 3. Loss of balance 4. Fast phase of nystagmus AWAY from the lesion No proprioceptive deficits — that is what makes it peripheral. Common cause: THO .	Treat the underlying cause.
Lead poisoning	ROARING and DYSPHAGIA — laryngeal and pharyngeal paralysis. A distinctive equine presentation of lead toxicosis.	Calcium EDTA chelation.

RAPID-FIRE EQUINE NEUROLOGY RECALL

Symmetric ataxia, worse behind, young TB → **Wobblers type 1** → **myelogram**.

Asymmetric ataxia with muscle atrophy → **EPM** → ponazuril/diclazuril 28 d.

Ataxic foal under 1 year → **EDM** → **vitamin E**.

Weight loss with a good appetite + trembling → **EMND** → **vitamin E**, tail-head biopsy.

Traveling horse, fever, then hind-end paresis, xanthochromic CSF → **EHV-1**.

Poor tail tone + fecal incontinence → **cauda equina** → **poor prognosis**.

Fine tremors + hyperesthesia → **West Nile**.

Neutrophilic CSF in an encephalitic horse → **EEE**.

Foal, 2 wk–8 mo, trembling and weak → **shaker foal (toxico-infection)**.

Roaring plus dysphagia → **lead**.

6 | Lameness, the Foot & Musculoskeletal

6.1 Conditions of the Foot

Condition	Definition	Treatment
Quittor	Chronic infection of the COLLATERAL CARTILAGES of P3 , with draining tracts at the coronary band.	Surgical debridement.
Sidebone	OSSIFICATION of the collateral cartilages of P3 , from repeated concussion on the quarters of the hoof. Often an incidental finding.	Usually none.
Canker	Suppurative HYPERTROPHY of the frog and sole — proliferative, foul-smelling tissue. Note it is proliferative , unlike thrush which is destructive.	Debridement, topical therapy, dry environment.
Thrush	Malodorous, black, degenerative infection of the frog sulci from wet, dirty footing.	Copper sulfate and EXERCISE , plus hygiene.
Pedal osteitis	Demineralization of the solar margin of the distal phalanx , usually secondary to inflammation and chronic bruising.	Corrective shoeing, rest.
Distal phalanx fracture (non-articular)	Fracture of P3 not involving the joint surface.	Bar shoe and rest.
Club foot	Contracture of the distal interphalangeal joint . Classic: a Quarter Horse at 6 months . Congenital, or acquired from nutritional imbalance .	Trimming, toe extensions, oxytetracycline in foals, deep digital flexor tenotomy if severe.

6.2 Joints, Tendons & Sheaths

Term	Definition
Bog spavin	Idiopathic synovitis of the TIBIOTARSAL joint — soft, fluctuant distension of the hock.
Thoroughpin	Tarsal sheath tenosynovitis — effusion of the sheath of the lateral deep digital flexor tendon at the tarsus . An incidental finding and NOT painful .
Curb	Thickening or bowing of the plantar tarsal ligament , at the plantar aspect of the hock.
Windpuffs	Digital sheath tenosynovitis . Treatment: nothing — a cosmetic blemish in an older working horse.
Bowed tendon	Superficial digital flexor tendinitis — convex swelling of the palmar metacarpus.
Tendon sheath most prone to tenosynovitis	The digital flexor tendon sheath (DFTS) .
Rupture of the common digital extensor tendon	In foals: swelling of the tendon sheath at the DORSOLATERAL aspect of the carpus . Often bilateral, often with flexural deformity; usually a good prognosis with splinting.
Stifle pouches	The medial femorotibial and lateral femorotibial pouches do NOT communicate with each other. The femoropatellar pouch communicates with the medial femorotibial in most horses — which matters for joint injections and infection spread.

6.3 Developmental Orthopedic Disease

Condition	Key Facts
OCD — most common location	The DISTAL INTERMEDIATE RIDGE OF THE TIBIA (in the tarsocrural joint). Failure of endochondral ossification.

Condition	Key Facts
Subchondral bone cyst	Most commonly at the MEDIAL FEMORAL CONDYLE. The classic stem: a young Warmblood or Thoroughbred with stifle effusion and lameness. Part of the OCD/DOD complex from failed endochondral ossification.
Flexural deformities in foals	Limit exercise and reassess daily. Oxytetracycline (high dose IV) is used for congenital flexural deformities — it chelates calcium and transiently relaxes the musculotendinous unit.
Carpal valgus	Lateral deviation distal to the carpus. Treat with regular hoof rasping and periosteal stripping to encourage growth on the CONCAVE side. Transphyseal bridging for severe or late cases.
Rib fractures in foals	Ribs 3–6 at the COSTOCHONDRAL JUNCTION — a common birth injury and a cause of sudden death from cardiac or pulmonary laceration. Palpate and ultrasound the thorax of every dystocia foal.

7 | Ophthalmology

Topic	Detail
Equine recurrent uveitis (ERU, "moon blindness")	The leading cause of blindness in horses; immune-mediated, with <i>Leptospira</i> as a recognized trigger. Appaloosas are predisposed. Two common sequelae: CATARACTS and GLAUCOMA.
The pupil discriminator	Active GLAUCOMA → MYDRIATIC (dilated) pupil. Uveitis WITHOUT glaucoma → MIOTIC (constricted) pupil. A dilated pupil in a painful eye should make you measure intraocular pressure.
Glaucoma treatment in ERU	Timolol — a beta blocker acting on the ciliary body to decrease aqueous humor production. Dorzolamide — a carbonic anhydrase inhibitor that also decreases production. (Contrast: latanoprost increases outflow — but is contraindicated in uveitis.)
Corpora nigra cysts	Cystic granula iridica on the dorsal pupillary margin, causing head-shy or spooky behavior when they obstruct vision. Treat symptomatic cysts with TRANSCORNEAL LASER PHOTOCOAGULATION. The ruminant equivalent of the corpora nigra is the GRANULA IRIDICA — same structure, different name.
Cataracts	Most common cause in the horse: chronic uveitis.

8 | Dermatology

8.1 Infectious & Parasitic Skin Disease

Condition	Agent	Lesions	Treatment
Dermatophytosis (ringworm) Z	<i>Trichophyton equinum</i> or <i>T. mentagrophytes</i>	Circular areas of alopecia and crusting, often in the girth and saddle area. Spread by tack and grooming equipment.	Griseofulvin ; topical lime sulfur or enilconazole. Disinfect tack.
Dermatophilosis ("rain rot", "rain scald") Z	<i>Dermatophilus congolensis</i> — a gram-positive facultative anaerobic actinomycete	Crusted, matted "paintbrush" tufts along the dorsum and rump after prolonged wetting.	Cytology: "RAILROAD TRACK" chains of cocci — two parallel rows. Keep dry; penicillin if severe.
Bacterial folliculitis	Three organisms: <i>Corynebacterium</i> , <i>Staphylococcus pseudintermedius</i> , <i>Staphylococcus aureus</i>	Papules and pustules in the saddle/girth region.	Antibacterial shampoo; systemic antibiotics if deep.
Pigeon fever / ulcerative lymphangitis	<i>Corynebacterium pseudotuberculosis</i> — gram-positive (stains PURPLE), facultative intracellular	Two syndromes to name: 1. Pigeon fever — deep pectoral abscessation (also abdomen and groin), giving the "pigeon-breasted" look 2. Ulcerative lymphangitis — limb swelling with draining tracts; cellulitis indicates progression	Lance and drain abscesses; systemic antibiotics for the lymphangitic and internal forms. Fly control — <i>Musca/Stomoxys</i> transmit it.
Epizootic lymphangitis FAD	<i>Histoplasma capsulatum</i> var. <i>farciminosum</i> — a chronic dimorphic fungal disease	Multiple swollen papules on the forelimbs and pectoral region with purulent drainage ; ulcerative papules. Transmitted by <i>Stomoxys</i> flies.	Cytology: budding YEAST with a HALO, inside macrophages. Amphotericin B. A foreign animal disease of horses — not present in the US.
Cutaneous habronemiasis ("summer sores")	Three larvae: <i>Habronema muscae</i> , <i>H. microstoma</i> , <i>Draschia megastoma</i>	Non-healing, exuberant granulation tissue at wounds, the medial canthus, and the prepuce — where flies deposit larvae.	Three-part control: regular anthelmintics (ivermectin) , fly control , and manure removal . Hard to inject locally because of the sensitive sites involved.
Onchocerciasis	<i>Onchocerca cervicalis</i> ; <i>Culicoides</i> vector	A "BULL'S EYE" lesion in the CENTER of the forehead — the buzzword. Also ventral midline dermatitis and uveitis.	Ivermectin (expect a transient inflammatory reaction as microfilariae die).
Aural plaques	Ear papillomas — papillomavirus	White, flat, depigmented plaques on the inner pinna. Transmitted by BLACK FLIES. They last FOR LIFE.	Generally none; cosmetic. Imiquimod is used but painful.
New World screwworm R FAD	<i>Cochliomyia hominivorax</i>	Larvae feed on living tissue, burrowing into wounds. Eradicated from the US by sterile male release.	REPORTABLE. Any suspect case must be reported immediately.

8.2 Mites

Mite	Morphology	Lesions	Treatment
<i>Chorioptes equi</i> the most common mite of horses	Surface. LONG legs, SHORT pedicles.	PASTERNS DERMATITIS — distal limbs, especially in feathered breeds. Also perineum and tail head. Seen in WINTER in the northeastern US.	Topical sprays and dusts — it lives on the surface. Lime sulfur.
<i>Psoroptes ovis</i> R	Surface. Oval body, LONG segmented pedicles.	Pruritic papules at the base of the mane, tail, and forelock. Eliminated from the US and REPORTABLE.	Lime sulfur. Report.
<i>Sarcoptes</i> Z	BURROWING. Short legs, long unsegmented pedicles.	Severe pruritus and self-trauma of the head, neck, and shoulders. Rare in horses. Burrowing causes swelling, hyperemia, papules, and pustules.	Ivermectin — systemic therapy is required for a burrowing mite. Zoonotic.

AMITRAZ IS CONTRAINDICATED IN HORSES

Amitraz is a formamidine acaricide that **causes impaction colic and ataxia in horses** and can be fatal. It is used in cattle and dogs — never reach for it in a horse with mange.

8.3 Neoplasia

Tumor	Key Facts
Sarcoid	Caused by BOVINE PAPILLOMAVIRUS types 1 and 2. The most common equine skin tumor; locally aggressive, does not metastasize, notorious for recurrence after incomplete removal. Treat with intralesional CISPLATIN plus debulking.
Squamous cell carcinoma	Five sites: penis · sheath · lips · eyelids · eye (third eyelid and limbus). Unpigmented, sun-exposed skin and mucocutaneous junctions. Locally invasive with a real metastatic rate — treat aggressively and early.
Granulosa (theca) cell tumor	See §11. The ovarian tumor producing testosterone and stallion-like behavior.

9 | Infectious Disease

9.1 Viral Disease

Disease	Virus	Signs	Diagnosis & Control
Equine Infectious Anemia (EIA, "swamp fever") R	RNA virus — a lentivirus (retrovirus)	Most common lab finding: THROMBOCYTOPENIA. Fever, anemia, weight loss, ventral edema. Lifelong carrier state. Transmitted by biting flies (tabanids) and contaminated needles.	AGID — the COGGINS test (or cELISA). REPORTABLE ; positives are quarantined for life or euthanized. No treatment, no vaccine.
Equine Viral Arteritis (EVA)	Arterivirus	Five signs: fever · limb edema · abortion · conjunctivitis ("pink eye") · respiratory disease. A vasculitis.	STALLIONS are the permanent carriers — they shed in semen for life. Transmitted by SHIPPED SEMEN as well as by aerosol. Manage by vaccination (test and document status before vaccinating a colt).
EHV-1 and EHV-4 (equine rhinopneumonitis)	Alphaherpesvirus	Three syndromes from EHV-1: respiratory disease, ABORTION, and neurologic disease. EHV-4 is mainly respiratory. The classic abortion stem: "snotty" young horses, then abortion storms 4–5 months later in late-pregnant mares. Also causes interstitial pneumonia in the foal and THYMIC ATROPHY.	Vaccinate pregnant mares at 5, 7, and 9 months of gestation, and again close to parturition. Latency and recrudescence make eradication impossible.
EHV-3	Alphaherpesvirus	Equine coital exanthema. NO effect on fertility or breeding — the lesions resolve, leaving depigmented scars on the vulva, perineum, penis, and prepuce.	Sexual rest until lesions heal. Reassure the client.
EHV-2	Gammaherpesvirus	Associated with respiratory disease and fever.	Largely supportive.
EHV-5	Gammaherpesvirus	Equine multinodular pulmonary fibrosis.	Corticosteroids ± antivirals; guarded prognosis.
Equine influenza	RNA virus — orthomyxovirus	Explosive outbreaks of fever and harsh dry cough in naive populations. Damages respiratory epithelium → 3 weeks of rest per week of fever.	A carrier state exists. Vaccinate; antigenic drift requires updated strains.
African Horse Sickness FAD R	ORBIVIRUS (same genus as bluetongue)	Four signs: hydropericardium · fever · respiratory difficulty · supraorbital/periorbital edema. Mortality up to 90% in the pulmonary form.	Spread by CULICOIDES midges. Prevent by vaccination in endemic regions. A foreign animal disease — report immediately.

9.2 Bacterial Disease

Disease	Organism	Features	Diagnosis & Treatment
Strangles	<i>Streptococcus equi</i> subsp. <i>equi</i>	Fever, nasal discharge, and abscessation of the submandibular and retropharyngeal lymph nodes. Guttural pouch empyema and chondroids create carriers.	Bloodwork: NEUTROPHILIC LEUKOCYTOSIS. Post-exposure immunity depends on local nasal antibody against the SeM PROTEIN — the basis for the intranasal vaccine.

Disease	Organism	Features	Diagnosis & Treatment
Purpura hemorrhagica	Immune sequela to <i>S. equi</i> (or vaccination)	A TYPE III hypersensitivity — immune complex vasculitis. Well-demarcated ventral edema and petechiae.	Three-part treatment: penicillin · NSAIDs (flunixin or phenylbutazone) · corticosteroids.
<i>Streptococcus equi</i> subsp. <i>zooepidemicus</i>	Opportunist	The most common cause of infectious infertility in horses (endometritis in the mare), and the main organism in pleuropneumonia.	Culture-directed antimicrobials.
<i>Rhodococcus equi</i>	Gram-positive pleomorphic intracellular coccobacillus	Foals around 1–4 months of age. Causes pyogranulomatous PNEUMONIA with abscesses, plus ulcerative enterocolitis and DIARRHEA. Insidious onset; foals compensate until they crash.	Gram-positive pleomorphic rods on a transtracheal wash plus culture; ultrasound for abscesses. Treat with a MACROLIDE — clarithromycin or erythromycin — PLUS RIFAMPIN. (Remember the mare-colitis risk.)
Potomac Horse Fever	<i>Neorickettsia risticii</i>	See §3.4.	Blood/fecal PCR; oxytetracycline.
Equine granulocytic anaplasmosis	<i>Anaplasma phagocytophilum</i>	Four signs: fever · limb edema · icterus · petechiae. Vector: <i>IXODES</i> tick.	MORULAE in NEUTROPHILS on a routine blood smear — diagnostic for only 3–5 days of the illness. Confirm with blood PCR. Treat with oxytetracycline plus supportive care — response is usually rapid, within 24–48 hours.
Lyme disease / borreliosis	<i>Borrelia burgdorferi</i> ; <i>Ixodes</i> vector	Three manifestations in horses: neuroborreliosis, uveitis, and cutaneous pseudolymphoma.	Serology (multiplex); doxycycline or oxytetracycline.
Equine piroplasmosis R	<i>Theileria (Babesia) equi</i> and <i>Babesia caballi</i>	PEAR-SHAPED bodies inside RBCs. Hemolytic anemia, fever, icterus, hemoglobinuria. Endemic in FLORIDA in the US.	Treat with IMIDOCARB and fluids; diminazene (Berenil) is an alternative. Reportable in the US.
MRSA Z	<i>Staphylococcus aureus</i>	Carried in the NASAL PASSAGES of horses and their handlers. A genuine hospital-biosecurity and zoonotic concern.	Culture and susceptibility; barrier nursing.
Glanders ("farcy") FAD R Z	<i>Burkholderia mallei</i>	Deep ulcers in the mouth/nasal septum and nodular ulceration of the skin with corded lymphatics ("farcy pipes").	EUTHANIZE. Zoonotic and reportable — a historical biowarfare agent. Do not attempt treatment.
Dourine FAD R	<i>Trypanosoma equiperdum</i>	A VENEREAL disease — the only trypanosome spread by coitus rather than a vector. "SILVER DOLLAR" skin plaques plus genital edema and neurologic signs.	REPORTABLE. Test and slaughter.
Contagious Equine Metritis (CEM) FAD R	<i>Taylorella equigenitalis</i>	Copious mucopurulent vulvar discharge and short-cycling in mares. STALLIONS are the asymptomatic carriers (on the external genitalia).	CONTACT THE STATE VETERINARIAN — that is the answer. Requires specific transport media and approved laboratories.

THE EQUINE FOREIGN ANIMAL DISEASES

CEM · epizootic lymphangitis · African horse sickness · glanders · dourine · screwworm. All are reportable and none is present in the US. Add **EIA** and **piroplasmosis**, which are reportable and *are* found here.

RAPID-FIRE EQUINE INFECTIOUS DISEASE RECALL

Coggins/AGID → **EIA** → thrombocytopenia → reportable, no treatment.

Stallion carrier + shipped semen + limb edema and abortion → **EVA**.

Abortion storm 4–5 months after a respiratory outbreak → **EHV-1** → vaccinate at 5, 7, 9 months.

Depigmented scars on the vulva, no fertility effect → **EHV-3 coital exanthema**.

Hydropericardium + *Culicoides* + Africa → **African horse sickness (orbivirus)**.

Foal with lung abscesses → ***Rhodococcus equi*** → **macrolide + rifampin**.

Morulae in neutrophils + *Ixodes* → **anaplasmosis** → oxytetracycline.

Pear-shaped bodies in RBCs → **piroplasmosis** → imidocarb.

Silver dollar plaques + venereal → **dourine** → report.

Deep oral ulcers + skin nodules, zoonotic → **glanders** → **euthanize**.

Mucopurulent discharge, stallion carrier → **CEM** → **call the state vet**.

10 | Parasitology

THE TWO PARASITES THAT MATTER MOST

Adult horses in North America: CYATHOSTOMES (small strongyles) — the number one parasite of concern today.

Young horses 2–5 months: PARASCARIS EQUORUM (ascarids) — the number one concern in that age group.

Nearly every equine parasitology question resolves to one of these two.

10.1 Nematodes

Parasite	Disease	Diagnosis & Treatment
Cyathostomes (small strongyles)	The #1 parasite of adult horses in North America. Larvae ENCYST in the gut wall and remain dormant for long periods, then emerge en masse in winter or early spring → larval cyathostominosis / TYPHLOCOLITIS with diarrhea, hypoproteinemia, weight loss, and colic. Necropsy: tiny gritty nodules in the CECUM and colon wall.	Moxidectin , or fenbendazole at the 5-day larvicidal dose . Add prednisolone to blunt the inflammatory response to mass larval death. Fecal egg counts do not reflect encysted burden — that is the whole clinical problem.
<i>Strongylus vulgaris</i> (large strongyle)	VERMINOUS ARTERITIS — larval migration through the CRANIAL MESENTERIC ARTERY causing thrombosis and arteritis → non-strangulating infarction and colic. Also a leading cause of peritonitis .	Ivermectin . (Milbemycin has a longer half-life.) Now rare because of modern anthelmintic use.
<i>Parascaris equorum</i> (ascarid / roundworm)	The #1 parasite of foals 2–5 months of age. Larval migration through the lungs causes " summer colds " / verminous pneumonia — nasal discharge and cough in a young foal. The danger: heavy burdens plus deworming → sudden mass die-off → SMALL INTESTINAL IMPACTION or RUPTURE . A well-known cause of post-deworming colic and sudden death.	FENBENDAZOLE for 5 days at 2–5 months of age — a benzimidazole is chosen deliberately because it is NOT a paralytic : the worms die slowly and pass gradually, reducing impaction risk. Pyrantel is an alternative. Do NOT use ivermectin in a heavily parasitized foal — rapid paralysis of a large worm mass causes obstruction. Ascarids have also developed macrocyclic lactone resistance.
<i>Dictyocaulus arnfieldi</i> (equine lungworm)	Found in the BRONCHI . Causes chronic cough in horses. The DONKEY is the definitive host and is usually asymptomatic — the classic history is a horse pastured with donkeys.	IVERMECTIN . Horses are typically dead-end hosts and do not pass larvae, so fecals are often negative — treat on suspicion.
Pinworms (<i>Oxyuris equi</i>)	Tail rubbing and perineal irritation.	Perianal tape test; ivermectin/pyrantel.

10.2 Cestodes

Parasite	Disease	Treatment
<i>Anoplocephala perfoliata</i>	ILEOCECAL IMPACTION and INTUSSUSCEPTION — it congregates at the ileocecal junction. A recognized cause of spasmodic colic. Oribatid mite intermediate host.	PRAZIQUANTEL (or a double dose of pyrantel). IVERMECTIN IS NOT EFFECTIVE AGAINST TAPEWORMS — it covers strongyles and heartworm but not cestodes. This is a favorite trick question.

10.3 Anthelmintic Pharmacology

Drug	Mechanism	Spectrum
Ivermectin	Increases release of GABA , the major inhibitory neurotransmitter, and opens glutamate-gated chloride channels → paralysis of the parasite.	Strongyles, ascarids (with resistance), <i>Dictyocaulus</i> , <i>Onchocerca</i> , bots, heartworm. NOT tapeworms.
Moxidectin	Increases neuronal inhibition via chloride ion channels and GABA. Longer half-life and better tissue persistence than ivermectin.	Encysted cyathostomes — its key advantage.
Pyrantel	Produces a DEPOLARIZING neuromuscular blockade.	Roundworms, hookworms, ascarids, pinworms; tapeworms at a double dose.
Fenbendazole	Benzimidazole — binds tubulin. Not a paralytic , so worms die slowly.	Ascarids in foals; 5-day larvicidal dose for encysted cyathostomes.
Praziquantel	Increases membrane permeability to calcium.	Tapeworms in horses and small animals.
Albendazole	Benzimidazole.	Nematodes and flukes in cattle and camelids.

MACROCYCLIC LACTONE RESISTANCE

The two **macrocyclic lactones** are **IVERMECTIN** and **MOXIDECTIN**. Two helminth groups have developed resistance to them: **small strongyles (cyathostomes)** and **ascarids (roundworms)**. This is why fecal egg count reduction testing and targeted, rather than rotational blanket, deworming is now standard.

11 | Reproduction & Neonatology

11.1 The Mare

Topic	Detail
Cycle	Seasonally polyestrous (long-day breeder), anestrus in winter. Estrus lasts 5–7 days; cycle length 21 days. Gestation 11 months (>330 days).
Follicle size for ovulation	35 mm or greater — the diameter at which hCG or deslorelin will reliably induce ovulation.
Two-week pregnancy on ultrasound	A small round anechoic structure with bright hyperechoic dashes on the dorsal and ventral aspects of the yolk sac — "SPECULAR REFLECTORS." Round shape at this stage is what distinguishes an embryonic vesicle from a uterine cyst.
Best time to evaluate cervical integrity	DIESTRUS — because the cervix is closed and tight under progesterone, so a defect is apparent. In estrus the cervix is normally relaxed and open.
Stage 2 parturition	20–30 minutes, during which the fetus is expelled. Equine parturition is explosive; anything longer is a dystocia and an emergency ("red bag," premature placental separation).
Most common cause of dystocia	POSTURAL abnormalities due to the foal's long extremities — not an oversized fetus. That distinction is the exam point.
Retained placenta	The part usually retained is the tip of the NON-GRAVID uterine horn , which is thinner-walled. Retention beyond 3 hours is an emergency in the mare — laminitis and metritis risk.
Metritis after breeding	A normal post-breeding inflammatory response resolves within 48 hours . Persistence beyond that defines the "susceptible mare."
Endometritis treatment	Three components: uterine lavage • antimicrobials • oxytocin (to promote clearance).
Intrauterine antibiotic	AMIKACIN is the only antibiotic labeled for intrauterine use in horses in the US and Canada. Extralabel options: ampicillin, ceftiofur, penicillin. Note aminoglycosides (amikacin, gentamicin, tobramycin) are ototoxic and nephrotoxic systemically.
Ascending bacterial placentitis	Three treatments: flunixin meglumine (anti-inflammatory/anti-prostaglandin), progesterone/altrenogest (myometrial quiescence), and broad-spectrum antibiotics . Classic sign: premature udder development and vulvar discharge.
Uterine artery rupture	Three-part management: hypertonic saline , aminocaproic acid (prevents clot breakdown), and — critically — KEEP THE MARE CALM AND QUIET . Excitement raises blood pressure and blows the clot. Do not over-resuscitate.
Granulosa (theca) cell tumor	The mare develops stallion-like behavior from high TESTOSTERONE . Hormones elevated: estrogen, INHIBIN, ANTI-MÜLLERIAN HORMONE (AMH), and androgens including testosterone. AMH is the most sensitive single test. The contralateral ovary is small and inactive. Unilateral ovariectomy is curative.
Embryo retrieval	Perform 7–8 days post-ovulation — the equine embryo enters the uterus late (day 6).
Persistent hymen	In cattle and horses, caused by segmental aplasia of the paramesonephric (Müllerian) ducts .
Twin reduction	Best performed early — around days 14–16 , before fixation, by manual crush.

11.2 The Stallion & Infertility

Question	Answer
#1 cause of infertility — STALLION	Testicular degeneration.
#1 cause of infertility — MARE	Endometritis caused by <i>Streptococcus equi</i> subsp. <i>zooepidemicus</i> — also the most common cause of infectious infertility overall.
Carrier states to remember	Stallions carry EVA (in semen, for life) and CEM (on the external genitalia).

Question	Answer
Priapism	Persistent erection without sexual stimulation that does not resolve — classically after acepromazine , and potentially permanent in stallions.

11.3 The Foal

Topic	Detail
Failure of passive transfer (FPT)	IgG < 800 mg/dL. (Partial FPT 400–800; complete FPT <400 in most references.) The equine placenta is epitheliochorial and diffuse , so no antibody crosses in utero — the foal is entirely dependent on colostrum.
Colostrum requirement	1.5–2 L for a 50 kg foal in the first 18 hours of life. Gut closure occurs by about 24 hours.
Most common cause of neonatal septicemia	<i>E. coli</i> . Almost always on a background of FPT.
Neonatal resuscitation	Epinephrine: 0.01 mg/kg — roughly 0.5–1 mL of the 1:1000 solution for a 100 lb foal, every 3 minutes. 1:1000 means 1 gram in 1000 mL = 1 mg/mL.
Doxapram	A CNS respiratory stimulant — used in dummy foal syndrome (neonatal maladjustment) when the foal is hypoventilating.
Rib fractures	Ribs 3–6 at the costochondral junction. Palpate and ultrasound every dystocia foal before handling it roughly.
Oxytetracycline in foals	High-dose IV oxytetracycline for congenital flexural limb deformities .
Oral iron	Causes liver injury and hepatic encephalopathy in foals — never give it.

RAPID-FIRE REPRO & NEONATE RECALL

Specular reflectors on a round vesicle → **~14-day pregnancy**.

Evaluate the cervix in **diestrus**.

Stage 2 lasts **20–30 minutes** — longer is an emergency.

Dystocia is **postural**, not fetal oversize.

Retained placenta → **tip of the non-gravid horn**.

Mare acting like a stallion → **granulosa cell tumor** → AMH, inhibin, testosterone.

Uterine artery rupture → **keep her calm**.

Intrauterine antibiotic → **amikacin**.

FPT → **IgG <800 mg/dL**; give **1.5–2 L colostrum** in 18 hours.

Septic foal → *E. coli*.

12 | Toxicology

THE EQUINE TOXICOLOGY SHORTLIST

Horses are uniquely or especially sensitive to a handful of toxins. If a toxicology question specifies a horse, run this list first:
ionophores · blister beetles · black walnut · fumonisin · yellow star thistle · red maple · fescue · bracken fern.

12.1 Toxins Where the Horse Is the Sensitive Species

Toxin	Source & Mechanism	Signs	Management
Ionophores <i>monensin, lasalocid, salinomycin</i>	HORSES ARE UNIQUELY SENSITIVE. Used in cattle as a coccidiostat and growth promoter; horses are poisoned by cattle feed. Ionophores disrupt cation transport across membranes.	MYOCARDIAL NECROSIS and subsequent DILATED CARDIOMYOPATHY . Colic, sweating, ataxia, sudden death, or delayed heart failure weeks later.	Determine prognosis with FRACTIONAL SHORTENING on echocardiography. Empty the GI tract three ways: mineral oil · activated charcoal · fluid therapy. No antidote.
Cantharidin (blister beetle)	Beetles baled into alfalfa hay . A potent vesicant that damages every epithelium it contacts.	Three signs: renal disease · hematuria and hemorrhagic cystitis · peracute death. Also colic and oral ulceration. The pathognomonic behavior: the horse DUNKS ITS MUZZLE in water to soothe oral pain. Causes cardiovascular and endotoxic shock.	Supportive; check calcium and magnesium, which fall dramatically. High mortality.
Black walnut	Shavings used as bedding.	LAMINITIS — acute, within 24 hours of exposure, with limb edema.	Remove the bedding immediately; cryotherapy, NSAIDs.
Fumonisin (<i>Fusarium</i> in moldy corn)	Moldy corn disease. Inhibits sphingolipid synthesis.	EQUINE LEUKOENCEPHALOMALACIA — liquefactive necrosis of cerebral white matter. Sudden blindness, circling, ataxia, death.	No treatment; remove the feed. Test suspect corn.
Yellow star thistle <i>Centaurea solstitialis</i> (and Russian knapweed)	Prolonged grazing, typically in California and the West.	NIGROPALLIDAL ENCEPHALOMALACIA — destroys two structures: the GLOBUS PALLIDUS and the SUBSTANTIA NIGRA . Two signs: TREMOR and DYSPHAGIA — the horse physically cannotprehend food and starves with a full feed tub.	PROGNOSIS: GRAVE. The lesion is irreversible. Euthanasia.
Sorghum / Sudan grass / Johnson grass	Cyanogenic and lathyrogenic grasses.	MYELOMALACIA of the LOWER spinal cord. Three signs: pelvic limb incoordination · urine dribbling (incontinence) · death. Also cystitis and ataxia.	Remove from the pasture. Damage is permanent.

Toxin	Source & Mechanism	Signs	Management
Bracken fern	Contains THIAMINASE (and ptaquiloside).	In HORSES: polioencephalomalacia from thiamine destruction — blindness, ataxia, seizures. (In cattle the same plant causes bone marrow suppression and enzootic hematuria — a different disease in a different species.)	THIAMINE supplementation in horses; remove the plant.
Red maple	WILTED leaves are the danger.	ICTERUS , methemoglobinemia, Heinz body hemolytic anemia, hemoglobinuria.	Supportive; transfusion. Remove access to fallen branches.
Slaframine	MOLDY RED CLOVER (<i>Rhizoctonia leguminicola</i>).	Hypersalivation — "slobbers." Otherwise benign.	Remove the clover. Reassure the client.
Snake envenomation	Pit vipers (rattlesnake, copperhead, cottonmouth). Horses are usually bitten on the muzzle while grazing.	COAGULOPATHY — the tested answer. Also massive local swelling, and the real emergency is airway obstruction from nasal/muzzle edema . Cardiotoxicity and rhabdomyolysis occur with heavy envenomation.	Antivenin if available; protect the airway — pass nasal tubes early. Fluids, analgesia, tetanus prophylaxis. Monitor coagulation and cardiac rhythm.
Perennial ryegrass staggers	Endophytic fungus in perennial ryegrass produces lolitrem B — a neurotoxicant that acts on GABA receptors .	Fine head tremors and nodding movements ; depression, ataxia, tremors. Worse when excited.	Remove from the pasture; usually reversible.
Fescue toxicosis	Endophyte-infected tall fescue produces ERGOPEPTINE ALKALOIDS , which are DOPAMINE AGONISTS .	In the MARE: PROLONGED GESTATION, agalactia, thickened placenta ("red bag"), dystocia, and abortion. Dopamine agonism suppresses prolactin → no milk. General signs: lameness, sloughing of the rear hooves, decreased weight gain, fat necrosis.	Remove pregnant mares from fescue 60–90 days before foaling. Treat with DOMPERIDONE — a dopamine ANTAGONIST — to restore prolactin. (See the correction below.)
Lead	Contaminated petroleum products or paint.	ROARING and DYSPHAGIA in horses (laryngeal/pharyngeal paralysis). General signs: acute blindness, dullness, recumbency, unawareness of surroundings.	Calcium EDTA to chelate, or succimer. Do NOT use sodium EDTA — it binds calcium and causes fatal hypocalcemia.

12.2 General Large Animal Toxins Relevant to Horses

Toxin	Key Facts
Pyrrolizidine alkaloids	Ragwort, fiddleneck (<i>Amsinckia</i>), groundsel, <i>Senecio</i>. Hepatic failure with megalocytosis, periportal fibrosis, bile duct hyperplasia ; secondary photosensitization. Look-alike for Theiler's disease . See §4.1.
Aflatoxin	Mycotoxin from <i>Aspergillus</i>, found in MOLDY GRAIN. The other major hepatotoxin.
Cardiac glycoside plants	Four to name: milkweed · gossypol (cotton seed) · OLEANDER · foxglove . Arrhythmias, sudden death.
Cyanide plants	Five: chokecherry · arrow grass · Sudan grass · Johnson grass · sorghum . Signs: bright red mucous membranes, bright red blood, bitter almond smell of the GI tract on necropsy . Blocks cellular respiration. Treat with sodium thiosulfate, sodium nitrite, and methylene blue. The other cause of cherry-red membranes is CARBON MONOXIDE. (Contrast: nitrate toxicosis gives BROWN blood.)

Toxin	Key Facts
Moldy sweet clover	Coumarin → dicoumarol , a vitamin K antagonist (warfarin-like) → spontaneous hemorrhage. Diagnose with PT (extrinsic pathway, factor VII).
Nightshade	Atropine-like — parasympatholytic . Treated with physostigmine , which itself causes dilated pupils, progressive paralysis, and depression at excess dose.
Water hemlock	The LEAVES (and roots) are toxic. Five outcomes: grand mal seizures · salivation · tachypnea · birth defects · death .
Oak / acorns	Tannins. Five signs: hemorrhagic diarrhea · abdominal pain · tenesmus · colic · death . In cattle the main lesion is renal damage.
Blue-green algae	Anabaena , Microcystis , Aphanizomenon . Post-synaptic depolarizing block; two pathways — nicotinic agonist and acetylcholinesterase inhibitor . Birds and cattle most affected, but lethal to any species drinking the water.
Ergotism	Claviceps purpurea , a parasitic fungus on rye , oat , wheat , and Kentucky bluegrass (and, unusually, on stored grain). Alkaloids cause vasoconstriction → thrombosis → gangrene of extremities.

13 | Pharmacology, Anesthesia & Clinical Numbers

Drug	Notes
Detomidine / xylazine	Alpha-2 agonists. Detomidine causes analgesia and HYPERGLYCEMIA (alpha-2 stimulation inhibits insulin release). Profound sedation with a head-down posture; watch for bradycardia and AV block.
Acepromazine	Phenothiazine — antagonizes central dopamine receptors. Hypotension; PRIAPISM in stallions , potentially permanent.
Dantrolene	Ryanodine receptor antagonist — post-anesthetic myositis and exertional rhabdomyolysis in horses; malignant hyperthermia in pigs.
Doxapram	CNS respiratory stimulant — dummy foal syndrome and neonatal resuscitation.
Buscopan	Antispasmodic and bronchodilator; used to confirm reversible airway obstruction and for spasmodic colic and rectal exam relaxation.
Clenbuterol	Beta-2 agonist bronchodilator for equine asthma.
Erythromycin / macrolides	Treatment for <i>Rhodococcus</i> in foals — but FATAL COLITIS in adult horses , including the treated foal's dam.
Doxycycline	Oral doxycycline is acceptable in horses. IV doxycycline — even at small doses — causes CARDIAC ARRHYTHMIAS, COLLAPSE, and DEATH. Never give it intravenously.
Polymyxin	Used as an anti-endotoxin agent. At high doses it is NEPHROTOXIC and NEUROTOXIC .
Aminoglycosides (amikacin, gentamicin, tobramycin)	Ototoxic and nephrotoxic. Amikacin is the labeled intrauterine drug.
Sulfadiazine / sulfamethoxazole	Antifolate therapy for EPM. Side effect: ANEMIA — monitor with a CBC every 2 weeks.
COX-2 selective NSAIDs	Firocoxib and meloxicam — spare gastric and renal prostaglandins.
Aminocaproic acid	Prevents breakdown of a formed clot — used to control excessive bleeding (uterine artery rupture, guttural pouch mycosis).
Tetracycline	Can cause primary photosensitization .

Clinical formulas

Formula	Detail
Mean arterial pressure	$\text{MAP} = \text{Diastolic} + (\text{Systolic} - \text{Diastolic}) / 3$ MAP is the pressure that determines tissue perfusion — keep it above 70 mmHg under anesthesia to prevent post-anesthetic myopathy.
Bicarbonate deficit	$\text{HCO}_3^- \text{ (mEq)} = 0.3 - 0.4 \times \text{BW(kg)} \times \text{base deficit}$ Example: a 470 kg horse with a base excess of -13 → $0.4 \times 470 \times 13 \approx 2,440 \text{ mEq}$.
Epinephrine 1:1000	1 gram in 1000 mL = 1 mg/mL.

Appendix A | Reportable & Foreign Animal Diseases

HOW THESE ARE TESTED

The exam rarely asks you to treat a reportable disease — it asks **what you do next**. The answer is **notify the State Animal Health Official or USDA Area Veterinarian in Charge, restrict movement, and do not treat**. If "contact the state veterinarian" is an option alongside a treatment, choose it.

Disease	Agent	Recognition	Action
Equine Infectious Anemia R	Lentivirus	Thrombocytopenia, fever, weight loss, ventral edema.	Coggins (AGID) ; lifelong quarantine or euthanasia.
Contagious Equine Metritis FAD	<i>Taylorella equigenitalis</i>	Mucopurulent vulvar discharge; stallion carriers.	Contact the state veterinarian.
African Horse Sickness FAD	Orbivirus	Hydropericardium, periorbital edema, respiratory distress; <i>Culicoides</i> .	Report; vaccinate in endemic areas.
Glanders FAD Z	<i>Burkholderia mallei</i>	Deep oral/nasal ulcers, farcy pipes.	Euthanize. Zoonotic; historical bioweapon.
Dourine FAD	<i>Trypanosoma equiperdum</i>	Venereal; silver dollar plaques, genital edema.	Report; test and slaughter.
Epizootic lymphangitis FAD	<i>Histoplasma farciminosum</i>	Pectoral/forelimb nodules with purulent drainage; yeast with a halo.	Report; not present in the US.
New World screwworm R	<i>Cochliomyia hominivorax</i>	Larvae in living tissue.	Report immediately.
Equine piroplasmosis R	<i>Theileria equi</i> , <i>Babesia caballi</i>	Pear-shaped intraerythrocytic bodies; hemolysis.	Report; imidocarb.
VEE R	Alphavirus	Encephalitis; horse is an amplifying host.	Report.
Rabies R Z	Lyssavirus	Any unexplained neurologic case.	Report; public health protocol.
Hendra virus FAD Z	Henipavirus	Australia; fruit bat reservoir; fatal to people.	Extreme PPE; report.
Vesicular stomatitis R Z	Vesiculovirus	Oral vesicles and erosions. Affects pigs, horses, and cattle — and is the vesicular disease that does affect horses, unlike FMD.	Report.

Appendix B | Numbers, Ages & Anatomy to Memorize

Number	Meaning
Gestation 11 months (>330 days)	Estrus 5–7 days; cycle 21 days; seasonally polyestrous.
Stage 2 parturition: 20–30 minutes	Longer is a dystocia emergency.
Follicle 35 mm	Size at which ovulation can be reliably induced.
Pregnancy detectable at ~14 days	Round anechoic vesicle with specular reflectors.
FPT: IgG < 800 mg/dL	Foal needs 1.5–2 L colostrum in the first 18 hours.
BUN:creatinine > 20:1	Prerenal azotemia.
Gastric capacity 8–10 L	Why gastric rupture happens.
Rectal tear grades 1–4	Grade 4 is full-thickness and an emergency.
Galvayne's groove appears at 10 years	On the upper corner incisor; reaches halfway by 15 and full length by 20.
Teeth: 36–44	Variable with canines and wolf teeth.
Ribs 3–6, costochondral junction	Foal rib fracture site.
Wobblers: C3–C4/C4–C5 (young), C5–C6/C6–C7 (older)	Dynamic vs. static compression.
Rhodococcus: foals ~1–4 months	Macrolide + rifampin.
Parascaris: foals 2–5 months	Fenbendazole 5 days.
C. perfringens type C: foals <10 days	Hemorrhagic diarrhea.
Shaker foal: 2 weeks – 8 months	Toxico-infectious botulism.
Pelvic flexure	Most common impaction site; cecum on the right.
Distal intermediate ridge of the tibia	Most common OCD site; subchondral cysts at the medial femoral condyle.